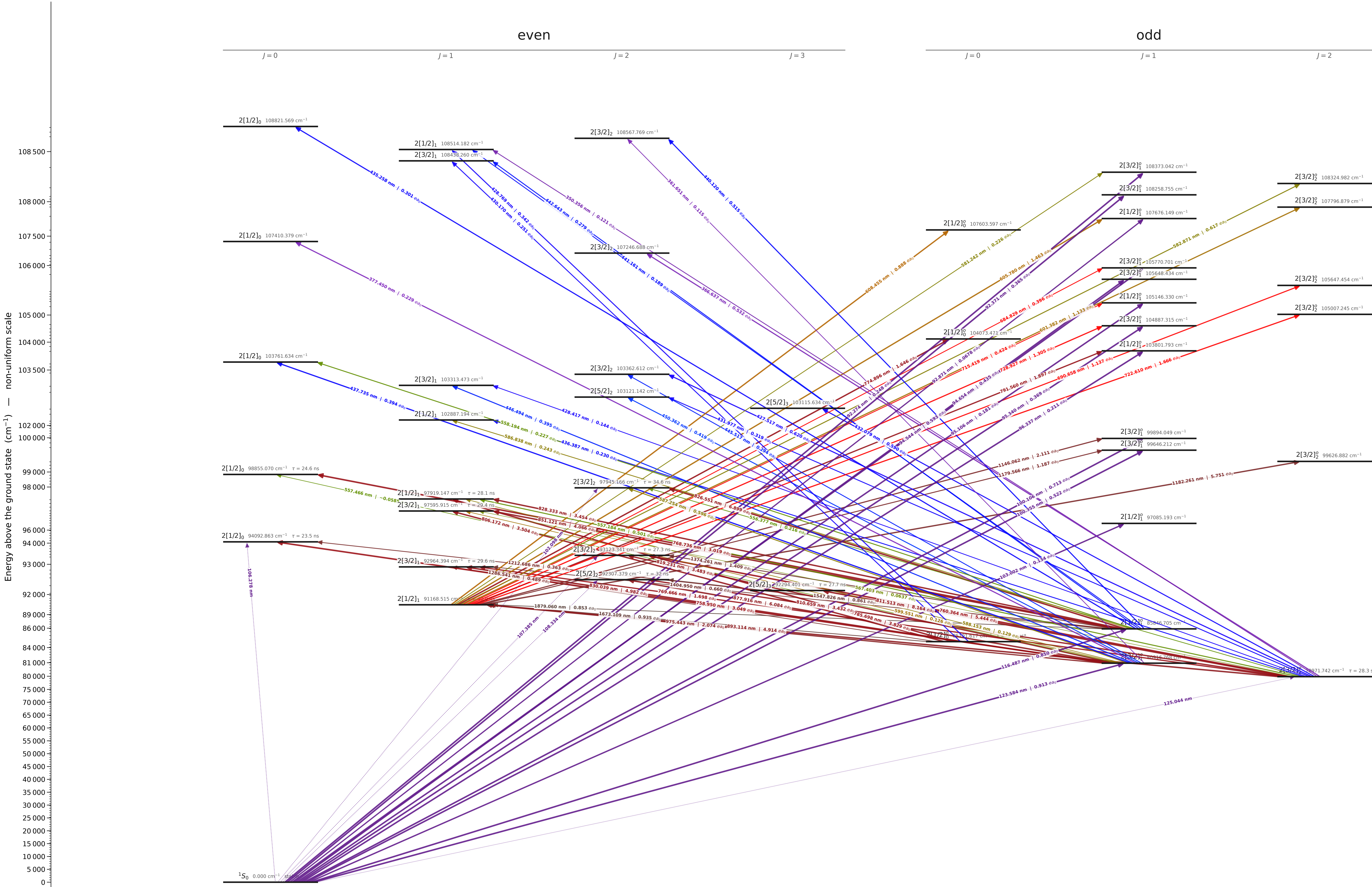
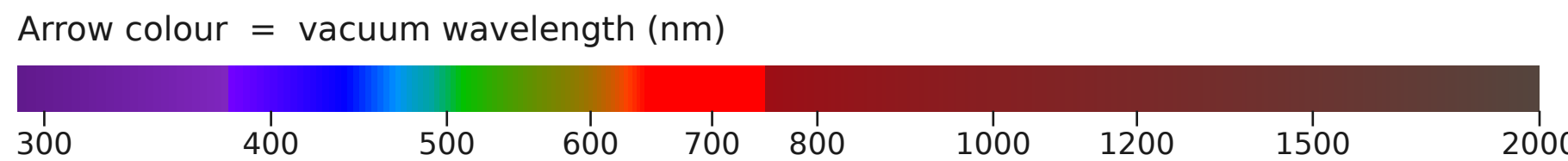


# Krypton

Kr (Kr I). Strongest classified transitions below 2 μm (NIST): 45 levels, 84 lines.

Kr<sup>+</sup> ionisation limit 112914.43 cm<sup>-1</sup> = 13.99961 eV (88.563 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J || e r || J' \rangle$  in ea0  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                        | λ vac (nm) | λ air (nm) | ν (THz)   | $\langle J    e r    J' \rangle$ (ea0) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|---------------------------------------------------|------------|------------|-----------|----------------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> | 96.3374    | -          | 3111.8995 | 0.211                                  | 3.350e+07            | -              | -          | 96.3 nm VUV line                                             |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.1061   | -          | 2994.7482 | 0.713                                  | 3.420e+08            | -              | -          | 100.1 nm VUV line                                            |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 102.0979   | -          | 2936.3222 | -                                      | -                    | 4.6            | -          | TALIF calibration line: two-photon excitation at 2 x 204...  |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.2780   | -          | 2820.8331 | -                                      | -                    | 6.773          | -          | two-photon line                                              |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.3845   | -          | 2791.7675 | -                                      | -                    | 5.82           | -          | two-photon excitation                                        |
| <sup>1</sup> S <sub>0</sub> - 2[5/2] <sub>2</sub> | 108.3337   | -          | 2767.3056 | -                                      | -                    | 4.967          | -          | two-photon line                                              |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 123.5838   | -          | 2425.8237 | 0.913                                  | 2.980e+08            | -              | -          | 123.6 nm resonance line: first                               |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 125.0442   | -          | 2397.4925 | -                                      | 3.540e-02            | 5.624e-09      | 100        | 125.0 nm M2 decay of the metastable 3P2 state                |
| 2[3/2] <sup>o</sup> - 2[3/2] <sub>2</sub>         | 587.2543   | 587.0916   | 510.4985  | 0.596                                  | 7.100e+05            | 4.6            | 2          | 587.1 nm fluorescence line detected in krypton TALIF         |
| 2[1/2] <sub>1</sub> - 2[1/2] <sup>o</sup>         | 605.7803   | 605.6126   | 494.8864  | 1.463                                  | 6.500e+06            | -              | -          | 605.78 nm line of 86Kr                                       |
| 2[3/2] <sup>o</sup> - 2[3/2] <sub>2</sub>         | 760.3638   | 760.1545   | 394.2750  | 5.444                                  | 2.732e+07            | 5.82           | 75.1       | 760.2 nm: decay channel that feeds the metastable 5s[3/2]... |
| 2[3/2] <sup>o</sup> - 2[5/2] <sub>3</sub>         | 811.5131   | 811.2900   | 369.4240  | 8.164                                  | 3.610e+07            | 5.747          | 100        | 811.3 nm                                                     |
| 2[3/2] <sup>o</sup> - 2[3/2] <sub>2</sub>         | 819.2308   | 819.0056   | 365.9438  | 3.483                                  | 8.940e+06            | 5.82           | 24.1       | 819 nm second step of the optical                            |
| 2[1/2] <sup>o</sup> - 2[3/2] <sub>2</sub>         | 826.5514   | 826.3242   | 362.7028  | 6.899                                  | 3.416e+07            | 4.6            | 97.7       | 826.3 nm: main decay branch                                  |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition                                        | λ vac (nm) | Isotopes | Shift (MHz) |
|---------------------------------------------------|------------|----------|-------------|
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> | 96.337     | 84-86    | -206(15)    |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> | 96.337     | 84-82    | 226(10)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> | 96.337     | 84-80    | 475(15)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> | 96.337     | 84-78    | 709(15)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.106    | 84-86    | -112(15)    |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.106    | 84-83    | 91(14)      |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.106    | 84-82    | 157(10)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.106    | 84-80    | 284(15)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> | 100.106    | 84-78    | 425(15)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.278    | 84-86    | -138(15)    |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.278    | 84-83    | 88(17)      |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.278    | 84-82    | 138(17)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.278    | 84-80    | 311(16)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sub>0</sub> | 106.278    | 84-78    | 426(15)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.385    | 84-86    | -130(15)    |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.385    | 84-83    | 83(16)      |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.385    | 84-82    | 148(13)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.385    | 84-80    | 295(14)     |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sub>2</sub> | 107.385    | 84-78    | 435(17)     |
| <sup>1</sup> S <sub>0</sub> - 2[5/2] <sub>2</sub> | 108.334    | 84-86    | -123(15)    |
| <sup>1</sup> S <sub>0</sub> - 2[5/2] <sub>2</sub> | 108.334    | 84-83    | 95(15)      |
| <sup>1</sup> S <sub>0</sub> - 2[5/2] <sub>2</sub> | 108.334    | 84-82    | 150(13)     |

Shift = ν(first isotope) - ν(second isotope).

## <sup>83</sup>Kr hyperfine structure (I = 9/2)

| Level               | A (MHz)      | B (MHz)    | Shift of each F level from the centre of gravity (MHz)                                                          |
|---------------------|--------------|------------|-----------------------------------------------------------------------------------------------------------------|
| 2[3/2] <sup>o</sup> | -244.04(3)   | -453.4(6)  | F=5/2: +2476.65 F=7/2: +1820.86 F=9/2: +864.35 F=11/2: -477.87 F=13/2: -2309.70                                 |
| 2[3/2] <sup>o</sup> | -160.500     | -105.900   | F=7/2: +834.21 F=9/2: +231.10 F=11/2: -748.73                                                                   |
| 2[1/2] <sub>1</sub> | -143.00(9)   | -          | F=7/2: +786.50 F=9/2: +143.00 F=11/2: -643.50                                                                   |
| 2[5/2] <sub>3</sub> | -103.910(20) | -437.0(7)  | F=3/2: +1514.21 F=5/2: +1345.48 F=7/2: +1083.77 F=9/2: +703.58 F=11/2: +172.14 F=13/2: -550.62 F=15/2: -1512.04 |
| 2[5/2] <sub>2</sub> | -156.49(8)   | -407.7(13) | F=5/2: +1534.53 F=7/2: +1165.18 F=9/2: +588.38 F=11/2: -272.31 F=13/2: -1510.34                                 |
| 2[3/2] <sub>2</sub> | -108.400     | -86.000    | F=5/2: +1152.98 F=7/2: +811.21 F=9/2: +350.28 F=11/2: -245.92 F=13/2: -997.10                                   |

Measured A, B constants; sources in the data file.